

Vocabulary: Causal Inference and ML - Week 10

Causal Discovery in Time Series

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Temporal Dynamics & Statistical Properties

- **Autocorrelation / Serial Correlation:** The linear or non-linear dependence of a variable's current value X_t on its own historical sequence $X_{t-\tau}$. Mathematically, the autocorrelation coefficient $\rho(\tau)$ at lag τ for a stationary process is defined as:

$$\rho(\tau) = \frac{\mathbb{E}[(X_t - \mu)(X_{t-\tau} - \mu)]}{\sigma^2}$$

where μ is the mean and σ^2 is the variance of X_t . *Violates the independent and identically distributed (i.i.d.) assumption, causing severe inflation of false-positive links if uncorrected.*

- **Stationarity (Weak):** The assumption that the statistical properties of a joint process remain invariant under time shifts. For a time series X_t , weak (or second-order) stationarity requires:

$$\mathbb{E}[X_t] = \mu, \quad \text{Var}(X_t) = \sigma^2 < \infty, \quad \text{Cov}(X_t, X_{t-\tau}) = \gamma(\tau)$$

meaning the mean and variance are constant, and the autocovariance depends solely on the lag τ , not on absolute time t . *In causal discovery, stationarity guarantees that the underlying structural mechanisms do not change, allowing repeated temporal relationships to be aggregated.*

- **Augmented Dickey-Fuller (ADF) Test:** A formal statistical hypothesis test used to determine if a time series possesses a unit root (is non-stationary). It estimates the regression model:

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{i=1}^p \delta_i \Delta X_{t-i} + \varepsilon_t$$

The null hypothesis $H_0 : \gamma = 0$ implies the process contains a unit root (is non-stationary). The alternative hypothesis $H_1 : \gamma < 0$ implies the process is stationary. *If a series is non-stationary, it must typically be transformed (e.g., via differencing $\Delta X_t = X_t - X_{t-1}$) to avoid spurious causal inferences.*

- **Granger Causality:** A predictive, time-precedence baseline definition of causality. Variable X Granger-causes Y if including the historical lags of X yields statistically superior forecasts of Y compared to utilizing the historical lags of Y alone. It evaluates the linear autoregressive formulation:

$$Y_t = \sum_{i=1}^p \alpha_i Y_{t-i} + \sum_{j=1}^q \beta_j X_{t-j} + \varepsilon_t$$

An F -test is performed to evaluate the joint significance of the coefficients under the null hypothesis $H_0 : \beta_1 = \beta_2 = \dots = \beta_q = 0$. *Strictly limited to linear modeling and Pearl's Level 1 (Association), as it relies entirely on passive observation without physical intervention.*

Time Series Graph Structures

- **Time Series Graph:** A directed acyclic or cyclic graph unrolled over time, where every node represents a distinct variable at a specific time step X_t^i , and edges capture directional temporal dependencies.
- **Lagged Causal Link:** A directed relation indicating that a variable's past state causes another variable's current state ($X_{t-\tau}^i \rightarrow X_t^j$ where $\tau \geq 1$). *Satisfies physical time precedence, making the direction of the arrow inherently identifiable.*
- **Contemporaneous Causal Link:** An instantaneous dependency occurring within the exact same time step ($X_t^i \rightarrow X_t^j$). *Typically emerges when the empirical measurement sampling rate is coarser than the true underlying physical process.*
- **Autoregressive Link:** A temporal causal link where a variable's past state directly influences its own future state ($X_{t-\tau}^i \rightarrow X_t^i$). *Explicitly models the internal memory or feedback loop of an isolated system component.*
- **Summary Graph:** An aggregated representation that collapses a time-extended graph into a variable-level representation by mapping repeated lagged edges onto timeless nodes.

Algorithmic Frameworks & Conditional Independence

- **FullCI:** A brute-force discovery framework that tests candidate links by conditioning on the entire past history of all variables in the system. *Suffers from the curse of dimensionality, becoming computationally expensive and statistically unstable when the number of lags or variables is large.*
- **PCMCI:** A data-efficient two-step time series causal discovery algorithm designed to handle high-dimensional autocorrelation. Phase 1 selects a compact set of candidate parents, and Phase 2 refines them using specialized independence testing.
- **Phase 1 (Condition Selection / Parent Selection):** An iterative PC-style pre-filtering step that finds a compact superset of the true parents ($\hat{PA}$) for each target variable, pruning irrelevant historical drivers. *Significantly reduces the size of downstream conditioning sets, preserving statistical power.*
- **Phase 2 (Momentary Conditional Independence - MCI):** A specialized testing step evaluating the link $X_{t-\tau}^i \rightarrow X_t^j$ by conditioning on the selected parent sets of *both* the source and the target nodes:

$$X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \hat{PA}(X_{t-\tau}^i), \hat{PA}(X_t^j)$$

Removes false positives driven by autocorrelation and indirect paths without overconditioning on the entire past.

- **PCMCI+:** An extension of the PCMCI framework explicitly designed to discover and orient contemporaneous links ($X_t^i \rightarrow X_t^j$) alongside lagged relations under time-stationarity.

Statistical Independence Tests & Implementation

- **ParCorr (Partial Correlation)**: A linear conditioning test that evaluates the strength of association between two variables while controlling for a continuous conditioning set. *Optimal for linear-Gaussian systems, but blind to non-linear dependencies.*
- **RobustParCorr**: A variation of partial correlation modified to downweight outliers. *Ensures the integrity of graph discovery workflows when facing heavy-tailed data distributions.*
- **GPDC (Gaussian Process Distance Correlation)**: A non-linear, continuous independence test that leverages kernel-based distance metrics to capture complex functional forms.
- **CMIknn (Conditional Mutual Information via k-NN)**: A completely non-parametric test that uses k -nearest neighbors estimation to quantify information transfer. *Highly expressive and free of functional assumptions, but demands substantial sample sizes.*
- **Lag Function Plot**: A visualization mapping estimated dependency strengths across a spectrum of time lags ($\tau = 1, 2, \dots, \tau_{\max}$). *Acts as an empirical diagnostic tool to verify optimal lag windows and identify fading causal impacts.*

Common Time Series Shortnames

Shortname	Meaning
ADF	Augmented Dickey-Fuller Test (Stationarity Check)
i.i.d.	Independent and Identically Distributed
FullCI	Full Conditional Independence Framework
PCMCI	Parent selection + Momentary Conditional Independence algorithm
MCI	Momentary Conditional Independence test statistic
PCMCI+	Extension of PCMCI that handles contemporaneous edges
LPCMCI	Latent-variable PCMCI (allows for unobserved temporal confounding)
ParCorr	Partial Correlation independence test
GPDC	Gaussian Process Distance Correlation
CMIknn	Conditional Mutual Information based on k -Nearest Neighbors
τ	Lag parameter representing time delay ($\tau \geq 1$ for historical links)